

Bedanta Bandyopadhyay

Aspiring Electronics Engineer | ML Enthusiast | CAD Modeler | Product Visualizer

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SUMMARY

Results-oriented engineer merging a robust background in automotive CGI with emerging expertise in Deep Learning and IoT. Experienced in managing end-to-end client projects and delivering high-impact visual solutions for diverse brands. Adept at rapid skill acquisition and passionate about building intelligent, connected systems

EDUCATION

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| 01. | Bachelor of Technology in Electronics and Communication Engineering
Institute of Engineering & Management, Kolkata | <i>Expected Graduation date: 2027</i>
<i>CGPA: 8.86 (Till 5th Semester)</i> |
| 02. | Higher Secondary (ISC)
Ram Mohan Mission High School | <i>Percentage: 73.2%</i> |
| 03. | Secondary (ICSE)
Ram Mohan Mission High School | <i>Percentage: 94.5%</i> |

TECHNICAL SKILLS

TOOLS AND SOFTWARES:

General: Microsoft Office | Git

Circuit and Embedded: Cadence Virtuoso | LT Spice | MATLAB | HFSS | Scaps1D | Arduino | Raspberry Pi | Jetson

PCB Design: KiCAD

Artificial Intelligence: Scikit-Learn | LangChain | RAG | HuggingFace

Database Management: SQL | MongoDB

CAD Engineering: AutoCAD | SolidWorks | Fusion360

3D Modelling: Blender3D | Autodesk Maya | Autodesk 3DS Max

3D Rendering: Blender Cycles | Chaos Corona Renderer | Chaos V-Ray | Luxion Keyshot

Environment Design: Gaea

Postproduction and Retouching: Adobe Photoshop | Blackmagic DaVinci Resolve | Blackmagic Fusion Design | Foundry Nuke

2D Illustrations: Adobe Illustrator

VFX: Adobe Premier Pro | Adobe After Effects

Texturing: Adobe Substance Painter | Adobe Substance Designer

Cloud and API Architecture: AWS | Flask

Web Development: HTML | CSS | ReactJS | ExpressJS | NodeJS

PROGRAMMING LANGUAGES: Python | C/C++ | Java | Rust | JavaScript

CORE COMPETENCIES: Circuit Design | Circuit Simulation | Fluid Simulation | UI-UX | CAD Modeling | Database Management Systems | Signal Analysis | Electronic Troubleshooting

ACADEMIC PROJECTS

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| 01. | OccuNova (Database Management Deep Learning): Developed a cascading Deep Learning model that detects the Glaucoma using Fundus Scan Images. | <i>Jun '26</i> |
| 02.2. | OccuNova (Database Management Deep Learning): Developed a cascading Deep Learning model that detects the Glaucoma using Fundus Scan Images after cropping into the retinal region of the scan using a U-Net Architecture. <i>WebDev</i> : Designed a fully functional React Website for a mockup for high profile restaurants, with a functional backend, Firebase and RazorPay integration for handling reservations, orders, deliveries and enquiries. | <i>May '26</i> |
| 03. | A.E.G.I.S. Core (Embedded Systems IoT Machine Learning): Developed a smart Women Safety device that works on Rpi 5 and Mediapipe to detect potential stalkers and disrupt any filming device(s) while transferring the location data of the victim to the concerned authorities and people. | <i>Feb '26</i> |
| 04. | A.E.G.I.S. Core (Embedded Systems IoT Machine Learning): Developed a smart Women Safety device that works on Rpi 5 and Mediapipe to detect potential stalkers and disrupt any filming device(s) while transferring the location data of the victim to the concerned authorities and people. [ONGOING]A smart agricultural monitoring system utilizing sensors and Arduino for data analysis for real-time crop health optimization. [PAPER] | <i>Jan '26</i> |
| 05. | Infrastructure-Assassin (Database Management Cloud LLM Cascade): Developed a cascaded LLM assisted SaaS that targets harmful, outdated, and malicious files in a company's database. It also has RBAC (Role Based Access Control) to better manage the requests and stop unauthorized logins/ action confirmations from users. | <i>Jul '26</i> |

PREVIOUS EXPERIENCES

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| 01. | Automotive and Product Visualizer: (<i>Freelance</i>): Extensive 7+ years tenure in remote freelance visualization. Specialized in hyper-accurate 3D design and lighting for automotive and product sectors. | <i>Apr '19 - Aug '25</i> |
| 02. | Co-Lead: (<i>Ninja Fellowship</i>): Led a team of 3d artists and worked alongside them to create striking visuals for various brands for over 2 years. | <i>Nov '22 - Mar '25</i> |
| 03. | 3D Generalist and Environment Artist: (<i>The AutoFocus Studio</i>): Specializing in environment design and generalist work. Delivered high-fidelity visualizations and technical artwork for studio projects and clients alike. | <i>Aug '25 – Feb '26</i> |
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INTERNSHIPS

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| 01. | Internship on Advanced and Cleaner Materials in Solar Technology:
Conducted performance optimization of Tri-Layer Lead-free Carbon-based Perovskite Solar Cells using Machine Learning and crafted a full detailed report on the entire process. [PAPER] | <i>Jun '25 - Jul '25</i> |
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CERTIFICATIONS

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| 01. | Business Intelligence and Analytics: (<i>NPTEL, IIT Madras</i>): Teaches about various machine learning algorithms that are used by companies for prediction of consumer response and sales of products. [VIEW] | <i>Jun '25</i> |
| 02. | Joy of Computing with Python: (<i>NPTEL, IIT Ropar</i>): Teaches about the basics of python and goes up to the various advanced topics that form the foundation of data science and analysis. [VIEW] | <i>Nov '24</i> |
| 03. | Cloud Computing: (<i>NPTEL, IIT Kharagpur</i>): Teaches about Cloud Computing and how various aspects and types of services are used and availed by the firms. [VIEW] | <i>Jun '25</i> |
| 04. | Introduction to Internet of Things: (<i>Coursera, University of San Diego</i>): Teaches about the basics of Internet of Things and how we have arrived at the current technological standpoint in this field and demonstrates the use of Dragon board for various projects. [VIEW] | <i>Oct '24</i> |
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EXTRA-CURRICULAR

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| 01. | Graphics Lead: Iris The Photographic Society of UEMK | <i>Sep '24 - Dec '24</i> |
| 02. | Media Lead: Iris The Photographic Society of UEMK | <i>Dec '24 - Aug '25</i> |
| 03. | Tech Lead: Iris The Photographic Society of UEMK | <i>Dec '24 - Aug '25</i> |
| 04. | Graphics Lead: Computer Society of India UEM Student Chapter | <i>Mar '25 - Jan '26</i> |
| 05. | Manager: Computer Society of India UEM Student Chapter | <i>Aug '25 - Jan '26</i> |
| 06. | Vice President: Computer Society of India UEM Student Chapter | <i>Jan '26 - Present</i> |
| 07. | Class Representative | <i>Jan '26 - Present</i> |
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